NexusOS
AI Business Operating System

Part 03.0
Linear Issue WAL-13

Validation & Testing

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-13
Report title	Validation & Testing
Generated	2026-07-19
Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Validation-and-Testing.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Validation & Testing

Linear: WAL-13 **Reconstruction date:** 2026-07-19

Evidence Statement

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Objective

Validate tenant isolation and core schema integrity with automated tests against live PostgreSQL.

Tests at Part 03.0 landing (`838db64`)

`backend/tests/test_tenant_isolation.py` originally contained **five** tests:

- `test_tenant_a_cannot_see_tenant_b`
- `test_get_across_tenants_returns_not_found`
- `test_querying_tenant_model_without_company_fails_closed`
- `test_same_email_allowed_in_two_companies`
- `test_soft_delete_hides_but_keeps_row`

These match the five “Verified guarantees” listed in `docs/product/database.md`.

Later test expansion (Part 03.1 — not 03.0)

The same file was later expanded (+11 tests) for repository hardening (restore, list_deleted, update protections, pagination, sort allowlist, count/exists, bulk_get, hard_delete). That expansion is Part 03.1 evidence (`f521d38`).

CI

`838db64` introduced `.github/workflows/ci.yml` running tests against PostgreSQL. Part 03.1 later added Redis, mypy, and `alembic check` steps.

Repository evidence

- `git show 838db64:ai-business-os/backend/tests/test_tenant_isolation.py`
- Current `backend/tests/test_tenant_isolation.py` (includes later additions — distinguish carefully)
- `docs/product/database.md` § Verified guarantees
- `.github/workflows/ci.yml`

Outcome

Tenant-isolation invariants became executable regression guards for the enterprise schema.

Evidence limitations (shared)

• Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.

• Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.

• No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).

• Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.